

NYU, Tandon School of Engineering
Bridge to CS
Homework #14

Submission Instructions

1. For this assignment, you should turn in 1 file:
 2. Name your file `hw14.cpp`
 3. Pay special attention to the style of your code. Indent your code correctly, choose meaningful names for your variables, define constants where needed, choose the most appropriate control flow statements, break down your solutions by defining functions, etc.
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(Weiss 7.40): The following divide-and-conquer algorithm is proposed for finding the simultaneous maximum and minimum:

1. If there is one item, it is the maximum and minimum.
2. If there are two items, then compare them, and in one comparison you can find the maximum and minimum.
3. Otherwise, split the input into two halves, divided as evenly as possible (if N is odd, one of the two halves will have one more element than the other).
4. Recursively find the maximum and minimum of each half, and then in two additional comparisons produce the maximum and minimum for the entire problem.

Write a function with the following signature to solve the problem recursively:

`vector<int> MaxAndMin(vector<int> &data, int start, int end).`

The function should return a `vector<int>` of two elements, where `element[0]` is the minimum value and `element[1]` is the maximum value. Do **NOT** include a `main` function.